INFOSEC TAMIL PRESENTS

Getting Started in VAPT & Offensive Security

A complete beginner's guide: cybersecurity streams, certifications, what to learn, where to practice, and a one-page roadmap you can follow start to finish.

Subscribe for more practical, no-fluff cybersecurity guidance.

THE BASICS

What Is Cybersecurity?

Cybersecurity is the practice of protecting computers, networks, applications and data from unauthorized access, damage or theft. In simple words, it is the art and science of keeping digital systems safe from people who want to break in or misuse them.

THE CORE IDEA: THE CIA TRIAD

Confidentiality

Only the right people can access the data.

Integrity

The data stays accurate and untampered with.

Availability

Systems and data are there when needed.

Everything you'll learn, offense or defense, ultimately protects one of these three things.

FIND YOUR PATH

Different Streams, Which One Should You Pick?

Honestly? It's up to you. Try each stream below, if one doesn't excite you, move to the next. The right stream is simply the one you enjoy doing.

VAPT / Red Team

Learn networking, web and Linux, then practice on TryHackMe and HackTheBox. (This video's focus.)

SOC / Blue Team

Start with networking and log analysis, then learn SIEM tools like Splunk or Wazuh.

GRC

Learn frameworks like ISO 27001 and NIST, and sharpen your documentation skills.

DFIR

Understand file systems and memory, then practice with tools like Autopsy and Volatility.

Malware Analysis

Build strong programming basics, then learn tools like Ghidra to study malware behavior.

Cloud Security

Go deep on one cloud platform (AWS, Azure or GCP) and learn its native security controls.

Threat Intelligence

Follow researchers, read threat reports, and learn to track attacker behavior (TTPs).

A COMMON QUESTION

Is Certification a Must?

Yes.

Think of certifications as entry tickets.

They get you past the gate

Many recruiters and ATS systems filter resumes by certification before a human even looks.

They prove structured knowledge

A cert shows you've validated your skills, not just gained confidence from videos.

They give self-study direction

A certification goal turns scattered learning into a clear, time-bound plan.

They matter most early on

As experience builds, real project and work history start to carry more weight.

Caveat: a certificate alone won't make you job-ready, pair it with real hands-on practice.

Types of VAPT

Vulnerability Assessment & Penetration Testing isn't one thing, it branches across many targets.

- Web Application
- Network (Internal & External)
- Mobile (Android/iOS)
- Cloud
- API
- Blockchain / Smart Contract
- Thick Client Application
- IoT / Hardware
- Wireless
- Source Code Review (SAST)
- Social Engineering
- ICS / SCADA (OT Security)
- Red Team Assessment

BEFORE YOU START

What Laptop Do You Need?

Good news: you do not need an expensive gaming laptop. You need enough power to run a virtual machine comfortably.



RAM

16 GB recommended, 8 GB bare minimum. You'll run Kali Linux and target VMs side by side.



Processor

Any modern multi-core CPU (Intel i5/Ryzen 5 or better) with virtualization (VT-x/AMD-V) enabled in BIOS.



Storage

An SSD, strongly preferred over HDD. At least 256 GB free, 512 GB+ is more comfortable.



Operating System

Windows, macOS or Linux all work fine as your host, Kali Linux runs inside a VM on top.



Internet Connection

A stable connection for online labs like TryHackMe, HackTheBox and PortSwigger.



That's really it

A decent mid-range laptop from the last 4 to 5 years is almost always enough to begin.

HOW TO START

Your VAPT Learning Path

Six building blocks. Each one listed here is just a starting example, there is always more to explore.

STEP 1



Computers & Networking

- TCP/IP, ports & services
- Port forwarding, subnetting
- ...and plenty more beyond these

STEP 2



Web Fundamentals

- HTTP headers, requests & responses
- Status codes
- DNS basics

STEP 3



Linux Skills

- File permissions
- Core commands
- Navigating the file system

STEP 4



Programming

- Shell scripting & Python
- AI can write code, but you still need
- to understand what that code is doing

STEP 5



Security Fundamentals

- Encryption concepts
- Hashing concepts
- Why both matter for VAPT

STEP 6



Practice, Practice, Practice

- Apply it all hands-on
- Free practice platforms
- Theory alone won't cut it

RECOMMENDED RESOURCES

Books & Guides Worth Your Time

Links for these will be shared separately, search the exact names below.

OWASP Testing Guide

The industry-standard, free methodology for testing web application security end to end.

The Web Application Hacker's Handbook

By Dafydd Stuttard & Marcus Pinto. A paid book, but still one of the best deep dives into real-world web attacks.

Online Books: No Starch Press

A publisher known for excellent, practical security titles, Black Hat Python, The Hacker Playbook, Practical Malware Analysis and Nmap Network Scanning are great picks once you're past the basics.

Use AI as a Learning Partner

Use tools like ChatGPT or Claude to explain concepts, quiz yourself, or break down confusing write-ups, but always verify important technical details yourself.

PUT IT INTO PRACTICE

Practical Practice Platforms

Click any name below, every link is live in this slide.

TryHackMe ↗

Guided, beginner-friendly rooms. The best place to actually start.

HackTheBox ↗

Realistic, less-guided machines once you're past the basics.

PortSwigger Web Security Academy ↗

Free and exceptional for learning web vulnerabilities hands-on.

PentesterLab ↗

Bite-sized, exercise-driven lessons on specific web vulnerabilities.

CyberWarFare Labs ↗

Scenario-based red team and attack simulation labs.

crAPI ↗

OWASP's intentionally vulnerable API, built specifically to practice API VAPT.

WebGoat & CloudGoat ↗

Vulnerable-by-design practice apps: WebGoat for web, CloudGoat for cloud (AWS).

THE COMPLETE PICTURE

Your One-Page VAPT Roadmap

Everything in this video, start to finish, on one page. Save this slide.

01

Build the Fundamentals

- Networking, Web, Linux basics
- Programming + encryption/hashing

02

Get the Right Setup

- 16 GB RAM, virtualization enabled
- Kali Linux running in a VM

03

Practice on Guided Platforms

- TryHackMe, PortSwigger Academy
- PentesterLab, HackTheBox

04

Earn an Entry Certification

- Your ticket into the industry
- e.g., Security+, eJPT

05

Explore VAPT Types

- Web, network, mobile, cloud, API...
- Find the niche that excites you

06

Go Hands-On, Vulnerable Apps

- crAPI (API), WebGoat (web)
- CloudGoat (cloud)

07

Level Up: Practical Cert

- OSCP, PNPT, CRTP and similar
- Proves real, hands-on skill

08

Build Your Career, Keep Growing

- CTFs, bug bounty, portfolio
- Stay curious, never stop learning